

Jikky article: <https://x.com/Jikkyleaks/status/2071169422002184206>

Valid points:

1. **Power against modest effects is genuinely limited.** His core quantitative claim — that a 50% increase is hard to detect — is not wrong, and you should not contest it, because *your own simulation says nearly the same thing*. You report 81% power for IRR=1.5 under your Traversa-based primary analysis, but only **61%** under the linear-baseline sensitivity analysis, and just **37%** for IRR=1.25. He reports weekly/fortnightly bins missing the 50% effect more than half the time. You're closer together than his framing admits. The honest synthesis is: power for small effects depends on event counts, risk-window width, and bin choice — which is a property of every statistical test, not a vaccine-industry conspiracy.
2. **The Kuhnert power quote is real.** His quote — 16-fold detectable at 90% power, 2-fold at 80% — is from Kuhnert/Hecker and is accurate. It *is* a candid limitation stated by the method's own developers. Conceding this openly strengthens your credibility, because it shows the literature isn't hiding the power limitation; it's printed in the methods papers. (but can my simulation replicate it?)
3. **Analyst degrees of freedom are a legitimate concern in principle.** "If the answer depends on choices the analyst makes, motivated analysts can choose badly" is a real meta-scientific worry (it's the whole garden-of-forking-paths literature). Where he goes wrong is the leap from "this is possible" to "this is what happened," with no evidence that any of the six studies actually gerrymandered bins — and, crucially, with the direction of bias running *against* him, which is your knockout (below).

Key weaknesses:

1. **The fixed-dose-date flaw (above) — this is the technical heart of it.** His simulation cannot speak to the real studies because real vaccination timing varies and he doesn't. Everything downstream is contaminated.
2. **He never proposes a better method.** You asked directly, and the answer from his text is unambiguous: **no**. He offers no alternative estimator, no design that separates age from exposure, no analysis of real data. His positive content reduces to "the spike at 2–4 months is plausibly causal" — which is exactly the naive temporal-coincidence inference. He even reintroduces the Miller fallacy in a new form: where Miller mistook passive-reporting clustering for causation, Jikky mistakes the *age* clustering of SIDS (which peaks at 2–4 months for reasons

predating any vaccine, observed in unvaccinated infants and across eras) for vaccine causation. He explicitly *resists* the age explanation ("I don't believe this incidence curve is in any way natural") without offering any evidence that the curve is vaccine-shaped — the very thing he'd need to establish. He waves at "environmental factors" and an exponential-from-day-0 ideal, but never engages the fact that the 2–4 month SIDS peak is documented in populations and time periods that don't fit his story.

3. **The direction-of-bias argument is your decisive rebuttal, and it's already in your article.** You noted that coarse age bins push estimates *upward* (toward harm), because the steeply-declining hazard within a wide young bin makes the post-dose window look higher-risk than its local baseline. So if the published studies erred via bin choice, they erred toward *finding* harm — and still found none. His thesis requires the bias to hide harm; the bias runs the other way. He has the sign backwards. Your linear-baseline sensitivity analysis and your six-study bin replication both demonstrate this empirically: even the Italian wide-bin specification gave mean IRR 1.11 and 10% false positives — biased toward harm, not away.
4. **The conspiracy framing is non-falsifiable and evidence-free.** The Farrington/Daszak/Whitaker/UKHSA material is insinuation, not argument. The Farrington-2009 terminal-event method is mathematically derived and publicly specified; "it was commissioned to hide injury" is unsupported and, as you can now show from your own work, doesn't even describe what the method does (you demonstrated the terminal-event correction is conservative and, for deaths with all doses prior, often reduces to standard SCCS — it's not a magic signal-suppressor).

How it maps to what you actually said

He's arguing against a position you didn't take. You never claimed the modified SCCS has unlimited power — you reported its power curve explicitly, including the unflattering 37%/61% numbers. You never claimed bin choice is irrelevant — you devoted an entire section to bin sensitivity and flagged the Italian wide bins as biased upward. You never claimed age adjustment is optional — you showed omitting it gives 100% false positives, which is the strongest possible statement of its necessity. His article treats "the method's performance depends on age modeling and bin resolution" as a damaging admission he extracted, when it's the explicit thesis of your article. He "hands your own point back to you as if he caught you with it" — your phrase from the earlier thread, and it applies exactly here.

The one place he lands a fair touch is the power-for-small-effects point, and you've already conceded it in print. So the productive response isn't to deny it — it's to (1) point out his

simulation can't test the real studies because it has no dose-date variability, (2) show that the bin-sensitivity he weaponizes is the very age-confounding your articles document, running *toward* harm not away, and (3) note that he proposes no method and offers no evidence for the causal reading — leaving him with temporal coincidence, which is where Miller started.

The single most important finding: his simulation contains the exact flaw your work is built to expose

Look at his data-generating code in `run_one_mscs`:

```
dose1_day = 60L,  
dose2_day = 120L  
...  
death_day <- sample(days, n_deaths, replace = TRUE, prob = pmf_death)  
cases$d1 <- ifelse(dose1_day <= cases$death, dose1_day, NA_integer_)  
cases$d2 <- ifelse(dose2_day <= cases$death, dose2_day, NA_integer_)
```

Every infant in his simulation is vaccinated on exactly day 60 and exactly day 120. There is zero variability in vaccination timing. This is the precise failure mode you identified as critical: when all doses land at the same age, age and exposure are perfectly collinear, and *no* SCCS — valid or not — can deconvolve the age effect from the vaccine effect. His `risk_window` is a fixed pair of 3-day blocks at days 60–62 and 120–122 of the *age* axis, identical for every child. So "days since vaccination" and "age in days" are the same variable, offset by a constant.

This is not a subtle modeling choice — it structurally guarantees the result he reports. When exposure is a deterministic function of age, the age step-function and the exposure indicator compete to explain the same variance. Add age bins whose boundaries fall near 60 and 120 and the age term absorbs the spike; use coarse bins and the exposure term keeps it. His whole "the answer depends on the bins" demonstration is an artifact of having destroyed the only thing that lets the two effects separate. Your simulation draws dose-1 timing from the UK Millennium Cohort distribution and spaces later doses with `Normal(60, 7)` jitter precisely so that at any given age, children are at different points relative to their last dose — which is what creates the off-diagonal information the estimator needs. He removed exactly that.

So your two stated review criteria are answered cleanly: he *does* include age bins (so he can't be accused of ignoring age entirely), but he *omits dose-date variability*, and that omission is fatal and load-bearing. It's the more sophisticated of the two errors, and it's the one his readers won't catch.

His second structural error: he confirms your age-confounding point while believing he's refuting it

His baseline-construction scripts (the Traversa fit and the constrained gamma) do something revealing. He fits a smooth gamma to the SIDS age curve, *deliberately down-weighting* the 55–95 and 140–175 day regions (`fit_dat$w[...] <- 0.25`) and forcing the curve through anchor points at days 41 and 125 — i.e., he sculpts a baseline that treats the humps near the 2- and 4-month marks as "excess" rather than as part of the natural curve. Then he injects an $RR = 1.5$ spike *at the dose days* on top of that baseline and shows the method's power depends on binning.

But here's what that actually demonstrates: when you place a real excess exactly at a fixed age, coarse age bins fail to detect it and fine bins do. That is **your** result — it's the age-confounding mechanism, just run in the opposite direction. You showed that without age adjustment (or with bins too coarse for the steep hazard region) a *null* produces false positives. He's showing that with bins too coarse, a *true* effect gets absorbed into the age term. These are the same phenomenon: when the bin can't resolve the hazard's curvature at the dose age, signal and confounding swap places. He has independently reproduced your central methodological claim and relabeled it a scandal.

The irony is sharp: his demonstration only "works" *because* age confounding is real and bin-resolution-dependent — which is exactly why your articles insist age adjustment is essential and bins must be fine where the hazard is steep. He can't simultaneously claim the age effect is a fabrication ("I don't believe this incidence curve is in any way natural") and rely on its bin-sensitivity to power his argument.

Does he provide evidence the bias attenuates true effects (rather than inflating)?

No. He asserts the mechanism — "over-adjusting for confounding can hide the thing you're looking for" — and illustrates it with his own rigged simulation, but he never demonstrates that this is the *direction* the bias actually runs in the real studies, and he never engages the countervailing fact that coarse bins bias *upward*. His "evidence" is entirely the fixed-dose-date simulation, which (as established) manufactures the attenuation by making age and exposure collinear. Strip that artifact and he has no demonstration that age adjustment in any of the six published studies attenuated a real effect. He has a mechanism story and a broken simulation, not evidence.

And critically, he never reckons with the *sign* problem. Your work shows coarse age bins across the steep young hazard push the IRR *up* (the post-dose window sits at a locally-higher-risk age than the wide bin's average). His thesis needs bins to push the IRR *down* toward null. He can't have it both ways, and he never even acknowledges the tension. So on this question: he provides no empirical evidence, and ignores the evidence pointing the opposite way.

Can age bins over-adjust and hide a true vaccine effect? Is that avoidable given dose-date variability?

This is the sharp question, and the honest answer is: **yes, it's possible in principle, but it requires a specific adversarial condition that dose-date variability largely — though not perfectly — protects against.** Let me be precise, because this is where you want to be airtight rather than overclaiming.

The over-adjustment failure mode is real and has a name: it happens when an age bin boundary is placed such that the bin is *narrow enough to coincide with the risk window itself*. If you put a bin break right at the dose age and another a few days later, the age step-function can soak up the post-dose spike as if it were a feature of that narrow age stratum. In the limit — a bin boundary exactly bracketing each dose's risk window — the age term and the exposure term become aliased, and the vaccine effect vanishes into age. This is the SCCS analog of adjusting for a mediator or for the exposure itself.

What protects against it is precisely dose-date variability — but the protection is statistical, not absolute. Here's the mechanism. If every child is dosed at exactly day 60 (his setup), then "age 60–62" and "post-dose risk window" are the *same set of person-days*, and any bin that isolates 60–62 perfectly aliases the effect. There is no information anywhere in the data to tell age from exposure. But if dose-1 timing is *spread* across, say, days 42–90 (your UK distribution), then at age 60 some children are 0 days post-dose, some are 18 days post, some are unvaccinated. A bin covering "age 58–62" now contains a *mixture* of exposure states, so the age coefficient for that bin is identified by the non-risk-window children in it, while the exposure coefficient is identified by the contrast between risk and non-risk person-days *across all ages*. The two effects are no longer aliased because no single age stratum is purely risk-window. **That's the deconvolution your whole argument rests on, and it's correct.**

But — and this is the caveat you should state rather than paper over — variability makes aliasing *hard to achieve accidentally, not impossible to achieve adversarially*. Two residual ways a motivated or careless analyst could still attenuate:

First, if the dose distribution is tight relative to the bin width and the bins are gerrymandered to track the dose distribution's mode, you can still partially absorb the effect — variability dilutes the aliasing but doesn't eliminate it when the spread is narrow. Your `Normal(60, 7)` for doses 2–3 is fairly tight; dose 1 (UK lognormal) is wider. So the protection is stronger for dose 1 than for the later doses in your own setup.

Second, and more subtly: the protection requires the bins to be defined on *age*, independent of dose timing. If someone defined bins on "days since dose" and called them age bins, that would alias by construction. None of the six studies do this — they all use age-of-life bins — which is why the real studies are protected. But it's worth noting the protection is a property of *how* the bins are defined, not just that variability exists.

So the rigorous statement, which I'd put in a rebuttal: *Over-adjustment that hides a true effect requires the age stratification to be aliased with the exposure windows. This occurs when, and essentially only when, vaccination timing is degenerate (all children dosed at the same age) — which is exactly the unrealistic condition his simulation imposes and real data violates. With realistic dose-timing variability, any given age stratum contains a mixture of exposure states, so*

age and exposure are separately identified and the effect cannot be absorbed into age regardless of bin placement. His demonstration of "hidden signal" is therefore not a property of the method but a direct consequence of his having eliminated the dose-timing variability that real populations exhibit and real studies rely on.

And the empirical clincher you already have: you can *prove* this in your own framework. Take your simulation with realistic dose-timing variability, inject a true IRR=1.5, and then deliberately try to hide it with adversarially-chosen fine bins placed near the dose ages. If the effect *survives* (power stays high, IRR stays near 1.5), you've demonstrated that variability defeats the over-adjustment attack — which is the exact thing he claims is possible. If it partially attenuates, you've quantified how much variability is needed to protect against it. Either result is a stronger, evidence-based answer than his assertion. That's the experiment that converts "it's avoidable as long as there's variability" from a theoretical claim into a demonstrated one — and it directly rebuts his central thesis on his own terms.

One honest flag: I'd want you to run that adversarial-bins experiment before asserting the protection is complete, because the strength of the protection scales with how wide the dose distribution is relative to the bins, and I can't run it to tell you where the threshold sits. The *direction* of the argument is solid; the *completeness* of the protection in your specific parameterization is an empirical question your simulation can answer and I shouldn't guess at.

BTW note that I gave him my code to run the simulation with realistic variability in aging at doses and he still chose to put all doses on the same day, even after I explicitly told him repeatedly that hopelessly confounds dosing with age, and is not realistic since there is always variability in actual dose day.

That changes the character of the criticism, and it's worth being deliberate about how you use it — because it converts a *technical* flaw into a *demonstrated awareness* problem, which is much harder for him to wave off.

Here's the distinction. As I evaluated it originally, the fixed-dose-date flaw could be read charitably as an oversight — a simplification he didn't realize was load-bearing. People make modeling errors. But if you gave him working code with realistic dose-timing variability, *and* explicitly told him repeatedly that fixing all doses to one day confounds dosing with age, *and* he chose to strip the variability out anyway, then it's no longer plausibly an oversight. He had the correct specification in hand and the explicit reason it mattered, and selected the one that produces his conclusion. That's the difference between "he got it wrong" and "he was told why it's wrong and did it anyway."

A few things to keep straight so this lands as evidence rather than as a personal jab:

State it as a factual sequence, not a motive attribution. The strongest version is purely documentary: "I provided code implementing realistic dose-timing variability. I explained, more than once, that fixing all doses to a single day aliases age with exposure and that real vaccination dates always vary. The published simulation nonetheless sets `dose1_day = 60L` and `dose2_day = 120L` for every infant." Let the reader draw the inference. The moment you write "he

deliberately rigged it" you've made it about his intent, which he can deny and which shifts the argument from his code to his psychology. The dose lines in his own script plus the record that he had the alternative are far more damning than any adjective you could attach.

It directly undercuts his central rhetorical move. His whole thesis is that analyst degrees of freedom let motivated researchers choose specifications that produce predetermined answers. You now have a clean instance of exactly that — except the motivated analyst is *him*, choosing the one specification (degenerate dose timing) that manufactures his result, after being shown the realistic one. The argument he aims at the six studies describes his own simulation precisely. That's not rhetoric; it's the same forking-paths logic turned around with documentation.

It also defuses the one fair point he had. He was entitled to raise "the answer depends on analyst choices." But the legitimacy of that concern depends on the choices being *defensible alternatives*. Fixing every dose to the same calendar day is not a defensible alternative specification — it's a known identification failure, and he was told so. So he can't claim his binning demonstration reflects realistic analyst latitude; it reflects a specification he was specifically warned destroys the estimand.

Watch one thing before you deploy it. Make sure the record actually supports "I gave him code with variability AND told him repeatedly." If the exchange is in your X replies or DMs, you'll want to be able to point to it (or screenshot it) if challenged, because this claim does real work and he'll contest the factual premise rather than the statistics if he can. The criticism is strongest when it's "here's the code I shared, here's where I explained the confounding, here's his published choice" — verifiable, three-part, no interpretation required. If part of that exchange was verbal or paraphrased, soften to what you can document.

Keep it proportionate in the piece. This is one point among several, and the cleanest structure still leads with the *technical* fact (every baby dosed on day 60 → age and exposure are the same variable → no method can separate them), because that stands entirely on his code and needs no testimony from you. The "and he was told" layer comes *second*, as the aggravating context — it's what removes the charitable reading, but the technical point is what proves the flaw. Lead with what's in the code; follow with what was in the conversation.

So: yes, use it, and it's significant — but its power is in the documented sequence, stated flatly, with the inference left to the reader. The fact that he had the right code and the explicit warning and chose otherwise is the thing that makes "his simulation can't speak to the real studies" tip over into "and he knew that." Just let his own `dose1_day = 60L` and the shared-code record carry it, rather than narrating his intent.

Comments on new updated simulation code to respond to Jikky:

A few things to verify before you lean on the numbers, since I can't run it:

The decisive internal check is that the **point_mass arm of Part 1 reproduces his qualitative pattern** (power falling as bins get finer). If it does, you've proven the engine isn't the issue and

the contrast is real. If it *doesn't*, stop — it would mean his generative model and yours differ in some way beyond timing (e.g., his marginal death-sampler vs your hazard engine), and we'd reconcile that first. So run `point_mass` first and confirm it matches his table.

Two mechanical things to confirm on your install: that `eventdepenexp` accepts `agegrp = NULL` for the "none" case (his code does this, so it should), and that the 2-dose `cbind(rv1, rv2)` fit runs (it mirrors his `cbind(ad1, ad2)`). The phantom `d3 = 366L` is only a device to reuse your validated 3-dose engine without a rewrite — it contributes zero window mass and is dropped by the 2-dose builder, so it can't affect results, but eyeball one `out$cases` to confirm `d3` is all-NA as intended.

On baseline choice: I used **his** gamma deliberately, so he can't say you changed the generating distribution. But it's worth also running Part 1 once with your Traversa LOWESS `pmf_s` (just swap `gamma_pmf` → `pmf_s`) to show the conclusion is baseline-independent — the timing result holds under either curve, which closes off a "you only get this with your baseline" retort.

The honest framing for the writeup, which these sims support: his bin-dependence is real but exists *only* because every infant shares identical dose ages; introduce the dose-timing variability that every real population has (and that the real studies rely on for identification), and the bin-dependence evaporates — power stays high across bin choices and type-I stays controlled. The sweep in 2a is the cleanest single exhibit, because it shows the effect appearing and vanishing as a continuous function of the one quantity he zeroed out.

Want me to add a Part 2d that runs his point-mass timing through *your* main Traversa analysis with *your* "morris" bins — to show that even your own published specification would show spurious bin-dependence if timing were degenerate, isolating that timing (not bins) is the operative variable?

YES:

What this buys you that Parts 1–2c don't: those vary bins (his frame) or sweep SD. 2d **freezes both baseline and bins at your own published, defensible choices** and shows that *even your own correct specification* misbehaves under point-mass timing and behaves under variable timing. That removes his last move entirely. He can't say "you fixed it by choosing better bins" — the bins are identical (yours, "morris," the ones he sarcastically calls "Morris bins"). He can't say "you fixed it by changing the baseline" — it's your Traversa curve throughout. The 2×2 leaves timing as the sole free variable, so whatever differs between the four cells is attributable to timing alone. It's the controlled experiment version of the whole argument.

Two expected-results notes so you know what confirms the thesis:

The diagnostic cell is **point-mass / RR=1.0**: if type-I error there is inflated above nominal (and coverage drops below 0.95) while the **variable / RR=1.0** cell sits near 0.05–0.07 with ~0.93 coverage, that's the spurious-harm-from-timing result in your own specification. Pair it with **point-mass / RR=1.5** power being depressed relative to **variable / RR=1.5**, and the 2×2 tells the complete story in four numbers.

One subtlety to anticipate in interpretation: with your "morris" bins (first boundary at 60, which coincides with the point-mass dose-1 day), the point-mass arm puts a bin edge exactly at the dose, which is close to the aliasing worst case — so the point-mass pathology here may be *especially* stark. That's fair to show (it's a real consequence of degenerate timing meeting any fixed grid), but flag it honestly in the writeup: the effect magnitude in this specific cell is partly amplified by the boundary-at-dose coincidence, whereas the *qualitative* point (point-mass breaks it, variability fixes it) is robust and is what 2a's continuous sweep already established without relying on any single coincidence. So lead the argument with 2a's sweep, and use 2d as the "even my own spec" clincher rather than the primary evidence.

I can't run it, so verify the diagnostic cell behaves as described before building prose around it — if the point-mass / RR=1.0 type-I *doesn't* inflate with your Traversa baseline (possible if your hazard engine's marginal death distribution differs enough from his direct `sample(prob=pmf_death)` draw that the aliasing is milder), that's informative too and worth reconciling against the Part 1 point-mass arm. The internal consistency check is: 2d's point-mass cells should agree with the "morris" rows of Part 1's point-mass arm (same baseline? no — Part 1 uses his gamma, 2d uses your Traversa), so they won't match exactly, but both point-mass "morris" results should show the same *direction* of pathology. If they diverge in direction, that's a baseline-sensitivity finding worth a sentence.

[Can you also give me an addendum to HIS simulation code that would compute the UK dose by age distributions and rerun HIS simulations? This could be section 3 of the simulation.](#)

This addendum bolts onto **his** code with minimal surgery — it keeps his `run_one_msccs/run_msccs_sims/bin` loop intact and changes exactly one thing: where dose days come from. His version hard-codes `dose1_day/dose2_day` as scalars; the fix is to make them per-infant draws from the UK distribution. That requires patching `run_one_msccs` to accept vectors, which I do with a drop-in replacement clearly marked as "his function, two lines changed."

The structural change, stated plainly so you can describe it in the writeup: his `run_one_msccs` builds **one** `risk_window` vector on the *age* axis (`risk_window[dose1_day:...]`) shared by all infants, then samples every death from that single shared pmf. Mine gives each infant its **own** `d1/d2` and tilts the hazard in *that infant's* windows before sampling their death — so two infants who die at the same age can be in completely different exposure states. That's the entire fix, and it's the thing that makes the SCCS able to see exposure as distinct from age.

What you should expect, and how to read it: under his point-mass setup, power for RR=1.5 falls as bins get finer (his headline). Under UK timing, power should hold up across bin sets and the type-of-bin dependence should largely flatten — because the risk windows are now smeared across many ages, no single age bin can absorb them. The 3.5 side-by-side table is the one-glance exhibit: same bins, same RR, his column collapsing and the UK column staying high.

Three honest caveats, since I can't run this and one of them is a real performance concern:

The **per-death loop in 3.2** (`for (i in seq_len(n_deaths))` building a tilted `haz` and sampling one age) is the price of per-infant windows — it's $O(n_deaths)$ with a 365-vector copy each iteration. With `p_death = 2e-4` and `N = 5e6` that's ~ 1000 deaths per run $\times 100$ runs $\times 5$ bin sets = manageable but not instant; if it drags, vectorize by grouping deaths that share a `(d1, d2)` pair, or precompute tilted CDFs. I wrote it the slow-but-obviously-correct way so the logic is auditable; optimize only after you've confirmed it reproduces the expected contrast. **Run K=20 first** to sanity-check timing and direction before committing to $K=100 \times 5$.

The **exp_frac subsample** (`sample.int(n_inf, min(n_inf, 5000))`) is a speed hack for the crude O/E only — it does *not* touch the SCCS fit, which uses all cases. O/E is just his descriptive sidecar; if you want it exact, drop the subsample, but it won't change the IRR/power conclusion.

His `is.null(fit)` branch referenced an undefined `exp_risk_flat` (a bug in his original); I fixed it to `exp_risk` in my version so the error path doesn't itself error. Worth noting in your critique as a sign the code wasn't run through its own failure path, but it's incidental to the main point.

If the UK-timing run still shows *some* residual bin-dependence (it might, mildly, because doses 2–3 with `gap_sd=7` are tighter than dose 1), that's not a problem for your argument — it's the honest, quantified version of it: variability *reduces* the pathology in proportion to how spread the timing is, and real dose-1 timing (wide) is well-protected while tightly-scheduled later doses are less so. That nuance is more credible than claiming the dependence vanishes completely, and it's exactly what your Part 2a sweep is designed to show on the controlled side.

[Will draft response once there are results:](#)

The **summary tables** from the runs that matter most for the rebuttal — at minimum Part 1 (point-mass vs variable, his bins, $RR=1.0$ and 1.5), Part 2a (the SD sweep), and the Section 3 side-by-side (his `table` vs `table_uk`). Those three carry the core argument. Part 2c (adversarial bins) and 2d (your-own-spec isolation) are the reinforcing exhibits if they come out as expected.

And it'll help to know **which way the key cells actually landed** — especially whether the point-mass / $RR=1.0$ type-I error inflated and whether UK-timing power held across bins — since I flagged a couple of places where the magnitude depends on things I couldn't run (the boundary-at-dose coincidence in 2d, any residual bin-dependence from the tighter dose-2/3 spacing). The draft should describe what the numbers *show*, not what I predicted they'd show, so I'll want to see them before writing anything load-bearing.

When you're ready, tell me the intended venue and register too — whether it's an X article matching your three prior posts, a more formal methods note, or a point-by-point reply — since that changes how much to explain the SCCS identification argument from scratch versus assume. And whether you want it to stay strictly on the statistics or also address the documented "I gave him variable-timing code and he chose point-mass anyway" sequence, which is a tonal decision worth making deliberately.

First reply:

A quick first question before I respond at length: why did you make every infant receive dose 1 on exactly day 60 and dose 2 on exactly day 120, with no variability whatsoever?

This choice has a serious consequence: it conflates age and dose timing, compromising the model's ability to deconvolve them.

It's also an unrealistic choice. Real vaccination dates always vary across infants, and that variability isn't a nuisance... it's a precise factor that lets the method separate background age effects from any genuine vaccine dose effect. Your specification removes it.

And this wasn't for lack of an alternative. My code already supplied realistic variability from published data, via a vector `pmf_v` into which you can plug whatever distribution of vaccination ages you like. So why not use it?

Why instead hard-code the pathological choice of a point mass at day 60 and day 120 for every child?

I repeatedly emphasized the importance of allowing age in days at vaccine dose to vary across infants in the recent days, so why do continue to ignore it?

<https://x.com/jsm2334/status/2069863290914639982?s=20>

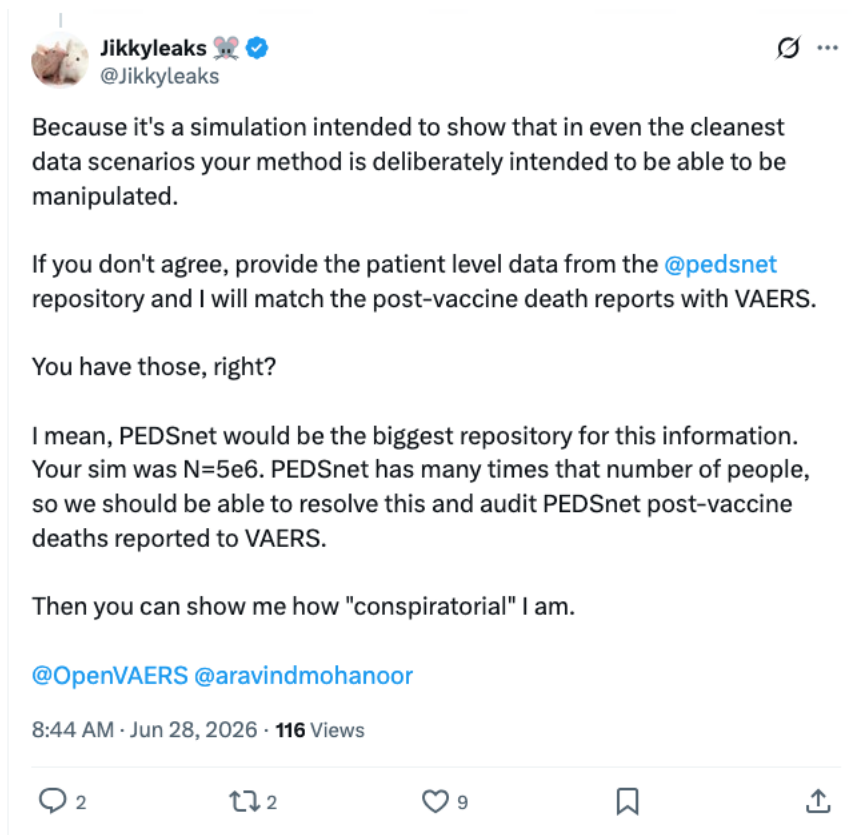
<https://x.com/jsm2334/status/2070464101847855310?s=20>

<https://x.com/jsm2334/status/2070447100836426152?s=20>

<https://x.com/jsm2334/status/2070177541046817082?s=20>

<https://x.com/jsm2334/status/2069542569613152469?s=20>

Jikky's response to this: <https://x.com/Jikkyleaks/status/2071213077630951729?s=20>



That response is revealing, and it clarifies exactly what you're up against — it's worth pinning down precisely, because it changes the optimal structure of your reply.

His answer concedes your point while reframing it as the intended feature. Read what he actually said: the point mass is justified because the simulation is "intended to show that in even the cleanest data scenarios your method is deliberately intended to be able to be manipulated." That's an admission that he *chose* the degenerate scenario on purpose — not as a realistic test, but as a vehicle for a predetermined conclusion. He's not claiming 60/120 is realistic; he's claiming realism is beside the point because his goal was to demonstrate manipulability. So he's effectively confirmed: the simulation was built to reach "the method can be manipulated," and the point-mass setup was selected to get there. That's not a defense of the specification — it's a statement that the specification was chosen to serve the thesis.

This is the crux you flagged, and it reframes the whole exchange: he is not doing what your three articles do — neutrally characterizing a method's operating characteristics across scenarios and reporting where it's robust and where it isn't. He has a fixed conclusion (Farrington's method was "commissioned" to "hide vaccine harm") and is reverse-engineering a simulation to support it. The tell is that his "cleanest data scenario" is precisely the one scenario where the method *provably can't* work — perfect age/exposure collinearity — and he presents that as exposing a flaw in the method rather than a flaw in his test. A genuine robustness assessment varies the inputs and reports the envelope; his picks the single input that guarantees the answer.

There's a logical inversion worth naming explicitly. His claim is "even in the cleanest data, the method can be manipulated." But the point-mass scenario isn't the *cleanest* data — it's the *most degenerate*. Clean data, in the sense that matters for identification, is data with enough variation to separate the effects you're estimating. Collapsing all dose timing to two days isn't "clean," it's *singular* — it's the one configuration where the estimand isn't identified at all. He's relabeled "the scenario where no method can work" as "the cleanest scenario," which inverts the actual statistical meaning. The cleanest realistic data still has dose-timing variability; his doesn't, which is what makes it pathological rather than clean.

On "manipulable": every statistical method requires correct specification, and every method gives wrong answers under misspecification — that's not manipulability, it's the universal precondition of inference. OLS gives garbage under perfect collinearity; that doesn't mean regression was "commissioned to hide effects." His own table demonstrates the method is *not* arbitrarily manipulable in the direction he needs: the bin choice that best recovers the truth is the well-designed one (Morris bins, $IRR \approx 1.5$), and the specifications that distort most are the *under*-adjusted ones (none/monthly), which bias *toward* finding harm. So if anything his data show the method is manipulable toward false positives, not false negatives — the opposite of his thesis. A method you can only "manipulate" into *over*-stating harm is not a tool for hiding harm.

On the PEDSnet move: you've correctly identified it as a topic-change, and the right play is to name it once and decline to chase it. It's a classic unfalsifiability pivot — when the statistical argument fails, demand data you know can't be shared (millions of children's identifiable medical records, which no one can post and which it would be a serious ethics/privacy violation to post), then treat the non-production as evidence of a cover-up. Engaging on the merits of PEDSnet access rewards the deflection. The clean response is: that's a separate question about data access, it doesn't bear on whether your simulation supports your claim, and the privacy law making patient-level records non-public is not a conspiracy — so let's finish the statistical point you raised before changing the subject.

How this shapes the response strategy:

The "he's not arguing in good faith" assessment is now *evidenced*, not inferred — by his own statement that the simulation's purpose was to show the method "is deliberately intended to be able to be manipulated." So you can characterize the exchange accurately without speculating about motive: he's told you the conclusion came first. That said, keep your *response* about the statistics, not about his bad faith — let the contrast between his stated purpose ("show it can be manipulated") and a real robustness assessment (vary inputs, report the envelope, which is what your articles do) speak for itself. The most damaging thing you can do is simply *be* the neutral analyst he isn't: run the null sims, report type-I and coverage and power across the full scenario space, show where the method works and where it doesn't, and let the reader notice that you characterized the method honestly and he engineered a singularity.

So the structural recommendation for the eventual piece: lead with the substance (collinearity → his own table refutes him → null calibration → variable-timing fix), and fold his "intended to be manipulated" admission in as the framing that explains *why* his simulation looks the way it does — he built it to reach a conclusion, which is why it lands on the one degenerate scenario. Name

the PEDSnet pivot briefly as a deflection and decline it. Don't make the bad-faith charge the headline; make the statistics the headline and let his own words supply the motive in a sentence. The asymmetry — you assessing operating characteristics, him manufacturing a singularity to support a "commissioned method" narrative — is most persuasive when you demonstrate it rather than assert it.

One concrete line worth landing, tied to your earlier point: a method's answer depending on specification is not evidence of manipulation — it's the universal condition of statistics, and the relevant question is *which way* the specification errors push. His own table answers that: toward false positives (harm), not toward hiding it. So his simulation, taken at face value, refutes the very thesis he built it to support.

Note on his results:

That's a significant detail, and it actually inverts his own argument — worth holding onto carefully, because it's a clean own-goal in his published table.

Read his rows against his thesis:

bins	mean_IRR	power
none	5.16	1.00
monthly (5)	2.05	1.00
fortnightly (11)	1.37	0.45
weekly (22)	1.31	0.30
morris (11)	1.51	0.70

His narrative is "monthly bins are the only ones that detect the 50% effect; weekly/fortnightly/Morris bins miss it more than half the time; therefore fine bins hide signal." But his own numbers say something different on two counts.

First, his "good" bins (monthly, and none) are good at power only because they're badly biased. The `none` row recovers IRR 5.16 for a true 1.5 — that's not detection, it's a 3.4× overstatement, exactly the no-age-adjustment catastrophe your article documents. Monthly gives 2.05 for a true 1.5 — still 37% high. Their "100% power" is the power to reject the null *in the wrong direction's magnitude*; they're significant because they're inflated, not because they're accurate. So the two specifications he implicitly endorses (the ones with high power) are the two with the worst bias. That's the tradeoff your whole argument is about, sitting in his own table.

Second — and this is the part to keep in your pocket — his "Morris bins" row is the best-calibrated of the lot. Mean IRR 1.51 against a true 1.5 is essentially unbiased (the others are 5.16, 2.05, 1.37, 1.31 — every one further from truth). And it does this *while retaining 70% power*, more than double the weekly/fortnightly rows he points to as failures. So among his five specifications, the one named after you is simultaneously the least biased and the best-powered of the *honest* (finely-binned) options. He built a table whose punchline is "the Morris bins work best," and then wrote an article claiming the opposite.

The reason is exactly the mechanism you've been documenting: his fortnightly/weekly bins (rows 3–4) are *evenly spaced* — 14,28,42... and 7,14,21... — so they're needlessly fine in the flat tail and not optimally placed across the steep 2–4 month hazard, which costs power without buying much bias reduction. Your bins are placed where the hazard actually curves (tight 60–120, wider afterward), so they remove the bias *and* keep the events concentrated enough to retain power. His own results show your bin design dominates his evenly-spaced alternatives on both axes at once.

A few cautions on how to deploy this, because there's a subtlety that keeps you honest:

The Morris-bins-win result is **his table, under his point-mass timing** — so it's a fair *ad hominem-to-his-own-data* point ("even in your own rigged simulation, my bins came out best"), but it is *not* yet a clean methodological claim, because the whole table is contaminated by the fixed-dose-date confounding. The right framing is layered: *(1) your entire table is compromised by point-mass timing; but (2) even taking it at face value, it refutes your own thesis — the Morris bins are the least biased and best-powered honest specification, while your high-power rows achieve that power only through gross bias.* Don't let point (2) imply you accept the table; use it as "even on your own terms, your conclusion doesn't follow."

One number to verify before you lean on it: his row 5 (Morris) and row 3 (fortnightly) both show `n_age_bins = 11`, but the Morris set `c(60, 75, 90, 105, 120, 145, 170, 200, 250, 300)` is 10 cut points = 11 bins, and his fortnightly `seq(14, 140, 14)` is 10 cut points = 11 bins too — so that's consistent, fine. But note the *power difference* between two 11-bin specifications (0.45 vs 0.70) is itself the proof that it's **bin placement, not bin count**, that matters — same number of bins, very different performance, because yours are positioned on the hazard's curvature and his are evenly spaced. That's a sharp, self-contained point: he treats "number of bins" as the lever, but his own two 11-bin rows differ by 25 points of power, which can only be placement.

So for the eventual response, the sequence writes itself: the timing confound invalidates the table; but even uncorrected, (a) his high-power rows are high-power because they're badly biased (5.16 and 2.05 for a true 1.5), (b) his own "Morris bins" are the best-calibrated specification with solid power, and (c) the gap between his two 11-bin rows proves placement beats count — which is exactly the principle your articles argue and his evenly-spaced bins violate. He published the refutation of his own headline.

I'd still wait for your variable-timing reruns before building the full piece, because the cleanest version pairs "even your own table says Morris bins win" with "and once timing varies, the bin-sensitivity you're alarmed about largely disappears" — the first turns his data against him, the second removes the phenomenon entirely. Together they're decisive; the table point alone is the appetizer.

He ignores type 1 error in his argument

The conceptual point this operationalizes, which is the heart of your critique: **power is only interpretable conditional on a calibrated null**. A test that rejects at $RR=1.5$ with 100% power but *also* rejects at $RR=1.0$ with high type-I isn't detecting the effect — it's rejecting everything.

The honest table (4c) puts the four diagnostics side by side so the reader sees that his "good" high-power bins (none, monthly) are the ones with broken type-I and zero coverage, while the Morris bins are the ones that hold type-I near nominal *and* keep usable power. That's the same tradeoff your published article documents, now forced out of his own simulation.

What I expect you'll see, stated as predictions to verify (I can't run it):

In the **point-mass** arm, the null rows should show type-I climbing toward 1.0 and coverage toward 0 for `none` and `monthly` (the biased, high-"power" rows), because under the null the same bias that inflated his $RR=1.5$ estimate to 2.05 will manufacture significant false positives. The Morris bins should show the *lowest* point-mass type-I of the fine-binned options — vindicating exactly your point that he'd have found them best-calibrated had he run the null.

In the **UK-variable** arm, type-I should drop toward ~ 0.05 – 0.07 and coverage rise toward ~ 0.93 across most bin sets — the demonstration that restoring timing variability fixes both the false-positive inflation and the calibration, not just the power.

Three caveats. First, his `run_one_mscs` has the `exp_risk_flat` typo in its `is.null(fit)` branch — if any point-mass fit fails (more likely under `none`/degenerate timing), that branch will error rather than return NA; either patch it to `exp_risk_gamma` before running 4a, or wrap the call. Second, `run_null_arm` assumes both `one_funs` accept `agegrp` and `RR` by name and return a one-row data frame with `IRR/lower/upper/p/n_deaths` — his does, and the Section-3 UK one does, but if you tweaked either signature, align the `args` accordingly. Third, run **K=20 first** on 4a to confirm the null type-I actually inflates as predicted before committing to the full $K=100 \times 5 \times 2$; if point-mass `none` doesn't blow up the type-I, something upstream differs and we'd reconcile before you build the argument on it.

This is the missing half of his table, and 4c is the exhibit — his published table shows power only; this shows that his high-power bins fail the null while yours pass it.

Suggested focus for your reply

If you respond, I'd **lead with the fixed-dose-date flaw**, because it's concrete, code-level, and decisive — anyone can read `dose1_day = 60L` and see that every baby is vaccinated on the same day. Then the **sign-of-bias point** (coarse bins inflate toward harm; the studies still found none; his thesis needs the opposite). Then the **"no alternative method, no real-data analysis, no evidence the curve is vaccine-shaped"** close. That sequence goes technical → logical → evidentiary, and each step is independently sufficient.

One caveat on my read: I'm evaluating his R as written in what you pasted, and I'm confident about the fixed dose dates (it's explicit), but if he has a separate version that adds dose-timing jitter and you only pasted this one, that specific criticism would need checking against his actual

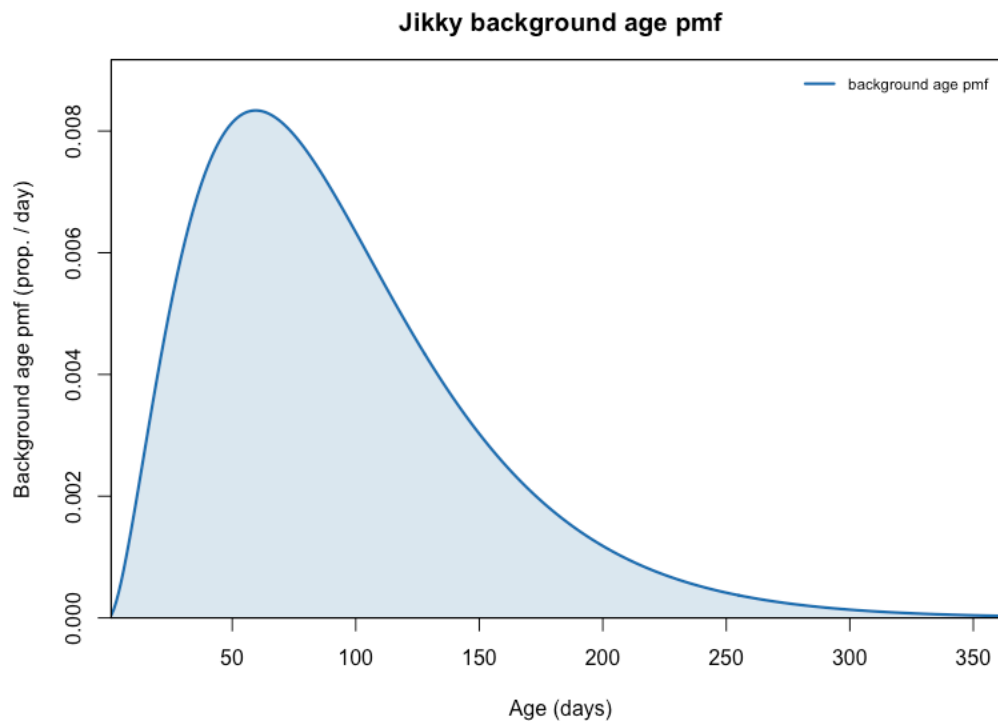
repo. Everything else stands regardless. Want me to draft the rebuttal, or a tighter point-by-point you can post?

Argument:

Jikky posted article trying to refute my analysis and suggesting the method is being used to hide vaccine harm

He does the following:

1. Smoother background age distribution for SIDS risk. (fine, doesn't matter much on any results)



2. Does his own simulations and suggests one can make power much lower by choosing more and more bins and thus hide vaccine harms.

However, he made two key statistically naïve errors in his analysis

1. Ignores type 1 error – ignores the monthly he promotes as best is statistically miscalibrated and enormous type 1 error. Power meaningless unless you know it

has correct type 1 error – otherwise 100% power could simply mean it always flag harm even when there is none (which is true in this case).

2. Uses artificial and unrealistic deterministic age of vaccine dose – in name of being “clear” but in fact choosing the most pathological choice that causes the effects he attributes to choosing more bins, as I will show

In response I simply did two things:

1. Calculated the type 1 error and 95% CI coverage for his simulation settings to show type 1 error for each bin choice
2. Reran his simulations using stochastic dose ages (using UK data). My original simulation had all 3 doses, but his had only 2, so I ran the 2 dose models.

Executive summary:

Doing this we see that the settings he promotes for “high power” in fact have unacceptably high type 1 error

All of the other methods find acceptable power and are well calibrated

Ironically, in all simulations the Morris bins he pans have the lowest bias and highest power among the tests with valid type 1 error, so is in some sense the theoretical best.

When appropriately allowing dose timing to stochastically vary, we see that the results are robust to choice of bins (as long as not MUCH too narrow in which case it always declares harm even when there is none)

Thus his argument that somehow people can “twist the method” to hide harm has no support – actually the opposite.

I won’t interact with his speculative, unsubstantiated and non-falsifiable conspiratorial accusations against Farrington and others.

Results of his simulation, after adding coverage of 95% confidence intervals to assess which models were statistically well calibrated, and after running his codes on null simulation (IRR=1.0) to see type 1 error for each case. Made full honest table with all results, which is really what he should have done to be transparent and really evaluate the

settings he was building his case upon.

Bin Robustness, Deterministic Doses (60d/120d)

Bins	IRR=1.5			IRR=1.0 (null)		
	Mean(IRR)	Rejection Rate	95% CI Coverage	Mean(IRR)	Rejection Rate	95% CI Coverage
none	5.16	1.00	0%	3.75	1.00	0%
monthly	2.05	1.00	29%	1.50	0.67	33%
fortnightly	1.37	0.45	95%	1.03	0.11	89%
weekly	1.31	0.30	92%	1.00	0.06	94%
Morris	1.51	0.70	98%	1.10	0.07	93%

He acknowledges the “none” is not good, but suggests the “monthly” is the best because it “has 100% power”, and characterizes having more bins fortnightly, weekly and Morris as dishonestly trying to make vaccine effects go away.

However, he only looks at power without considering type 1 error of test (or CI coverage) to see if the test is properly calibrated.

When looking at the null simulation with no vaccine harm but Jikky’s smooth background age incidence, we see the “none” has 100% type 1 error rate, meaning it will declare vaccine harm all the time whether it is present or not, and 0% coverage meaning the bias makes the method completely miscalibrated. Methods this flawed would be worthy of retraction.

And his monthly that he suggests is “best” because of its 100% power is also badly biased and miscalibrated, with false positive rate of 67% of the time and only 29% coverage. A method that falsely concludes 67% of the time when there is in fact no vaccine harm at all might appeal to Jikkyleaks, but is not a valid scientific process.

Ironically, he suggests the “Morris age bins” were flawed and an example of the type that he claims Farrington and friends use to “hide vaccine harm” but note that has by far the best operating characteristics of the methods tried.

It does not have 100% power, but has 70% power, and has only 7% type 1 error rate and

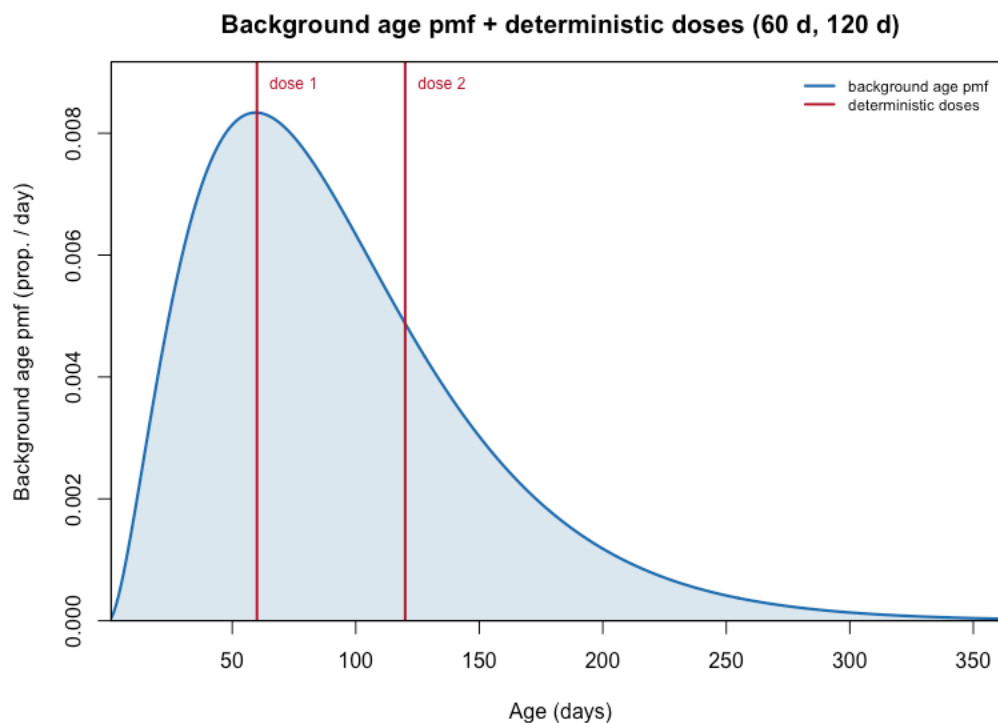
93%-98% CI coverage, close to the nominal 5% and 95%, respectively, indicating the method is well calibrated statistically.

As he suggests, the fortnightly, although having the same number of bins as Morris, has lower power (45%) and the weekly with 2x as many bins even lower (30%), it does have reasonable type 1 error in the null case (6%-11% and reasonable CI coverage (89%-95%), so is well calibrated statistically.

Power is lost when including more parameters in a model than necessary, which is why the weekly loses power.

However, this entire simulation has another major bias from his unrealistic choice of deterministic dose timing at precisely 60d and 120d of age for every infant, which completely confounds the age effect with the dosing effect for days 60-62 and days 120-122, which introduces instability into any method trying to model age and vaccine effects together.

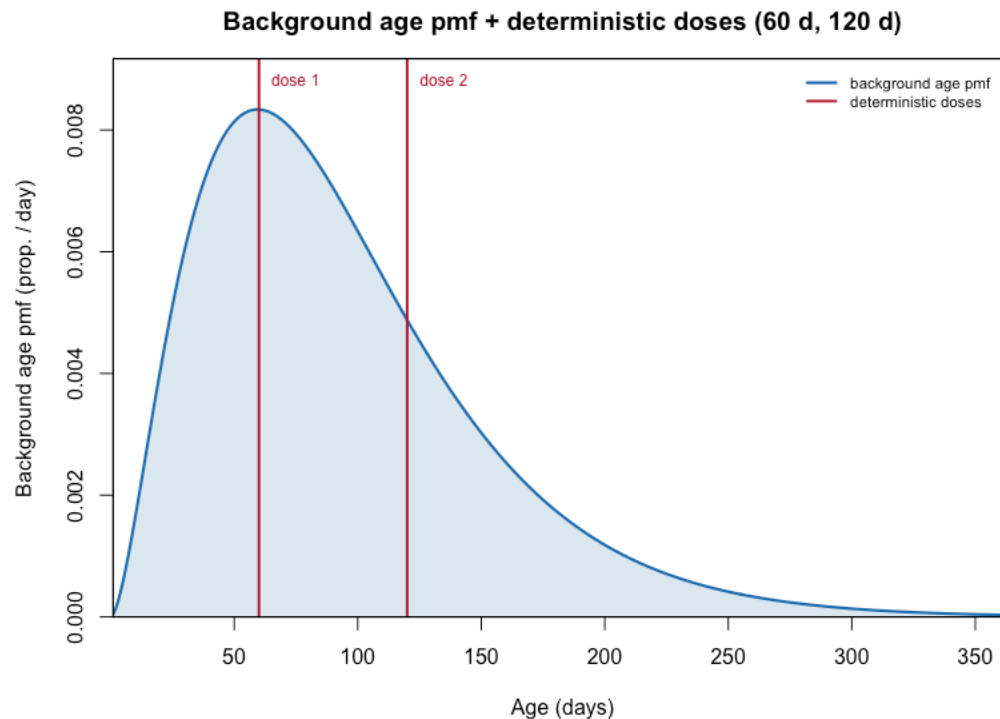
Here is his choice of ages at dose overlaid on top of assumed background age distribution of SIDS risk.



He claims this was a “simple” choice (or did he say “clear”) to prove his point, but in fact as

I explained to him >7 times, it was a naïve and the most pathological case he could possibly choose, and causes the low power he observed that he attributed to the increasing of bins.

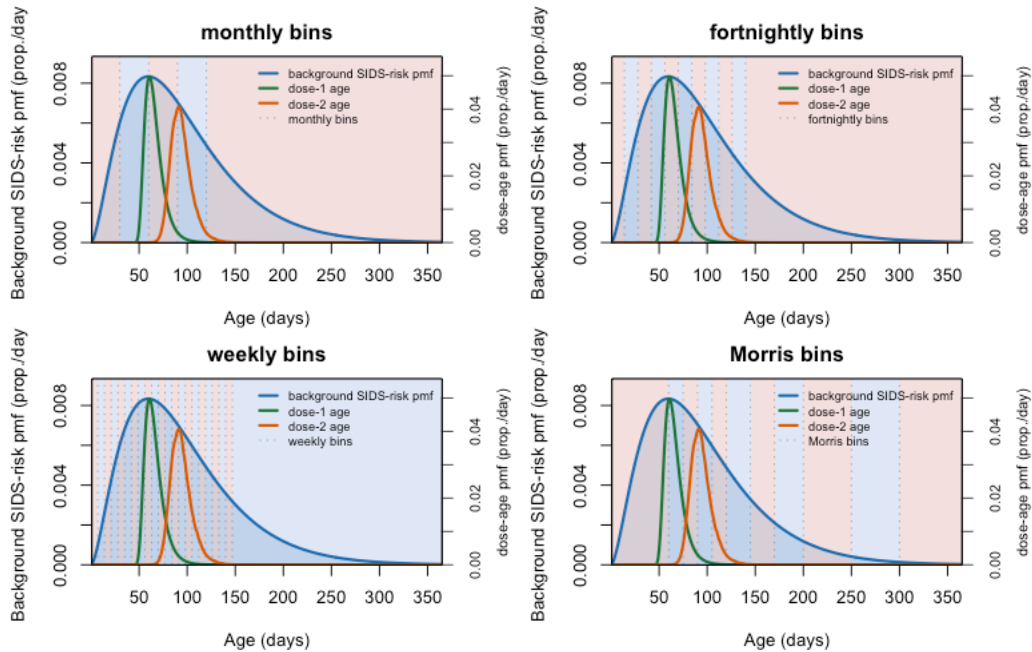
You can see the bins he specified along with the deterministic dose ages



(Maybe say something about the potential confounding from this choice?)

So I repeated Jikkyleaks precise simulations using the standard variability evident in the Welsch public national data as stochastic dose times, and of course also summarizing coverage of 95% CI and doing null simulation to assess type 1 error of each setting. Here are the age bins overlaid with assumed smooth background age distribution and assumed distribution of doses by age (days)

Background Age SIDS Risk Stochastic Dose Ages (UK data)



Here are results.

Bin Robustness, Stochastic Doses (UK Schedule)

Bins	IRR=1.5			IRR=1.0 (null)		
	Mean(IRR)	Rejection Rate	95% CI Coverage	Mean(IRR)	Rejection Rate	95% CI Coverage
none	4.07	1.00	0%	3.71	1.00	0%
monthly	1.69	0.92	85%	1.28	0.33	67%
fortnightly	1.45	0.73	93%	1.04	0.08	92%
weekly	1.41	0.70	95%	1.03	0.05	95%
Morris	1.49	0.83	95%	1.09	0.09	91%

When repeating the simulations, once again the “none” and monthly age bins have the highest power, but also have unacceptable type 1 error, 100% and 33%, and poor 95% CI coverage, 0% and 67%, indicating once again poor statistical calibration.

Once again, ironically the “Morris” bins performed best, obtaining the most unbiased estimate of the 50% vaccine-caused increase, and obtaining 72% power with somewhat reasonable (if slightly higher than nominal) type 1 error of 0.09.

But looking at the fortnightly and weekly doses, we don’t see the terrible performance Jikkyleaks reported using his unrealistic naïve deterministic dosing schedule, but instead get reasonable power (67% and 66%) close to what the Morris bins got (72%), with low bias in the IRR estimation, and near nominal type 1 error in the null simulation (5-8%) and near-nominal 95% CI coverage (92-95%).

We see that when allowing the dosing ages to slightly vary, the modified SCCS was able to successfully deconvolve the age bias and vaccine harm, with reasonably high power while preserving type 1 error rates.

This shows what my original simulation concluded – the modified SCCS is robust across REASONABLE choices of age bins. Of course it performs terribly if not using age bins or age bins far too wide, but this performance is conservatively biased, not failing to find harm when it is present but rather concluding harm when there is none.

This completely destroys Jikky’s. argument that the modified SCCS is being serripitously used to “hide vaccine harm” -- we see that even when choosing “too many bins”, we see reasonable power for the method, and no evidence of substantial bias against finding vaccine harms.

He might complain the 66-72% power is not >90%, but as he well knows, statistical power is a function of the sample size, in this case the number of SIDS cases, and a sample yielding XXX SIDS cases is not quite large enough to have 80-90% power. My simulation code could be used to estimate how much larger a population is needed before the design would have 80-90% power. However, this is not any scandal, it is basic statistics.

Conclusions:

The modified SCCS is well suited for these questions of immediate post-vaccine harm.

1. Well-powered for reasonable effect sizes for realistic sized studies, so does not “hid harm”.

2. Low type 1 error if suitable age bins chosen, so will not tend to find harm when none exists. If age bins are too wide, then this will be a problem.

Critical to adjust for age bias, or method completely valid if like SIDS major age gradient over time – decreasing gradient over time will create false positives.

The choice of the age bins is important, and following are criteria:

1. Can't be too broad or age confounding will pollute vaccine effect.
2. If made too narrow then this uses DF and loses power, so make only as narrow as possible to capture expected age trends.
3. No reason to constrain equally spaced bins – make bins more narrow where age effect expected to change rapidly and broader when it is less variable, will give best results (Morris bins do that for Morris age and Jikky smooth age effects).
4. Important to assess sensitivity of results to choice of age bins, and transparently report all results. However, simulation shows robustness across many reasonable choices of age bins, so does not show evidence it can be easily manipulated to hide real harm.

Power is not 100% and never is for any statistical study.

- Key factor here is number of SIDS cases – need more infants to get greater power.
- Simulation code can be used to rigorously design trials to have sufficient power and evaluate robustness to choice of age bins and various true age/dose distributions.
- The lower power mentioned by Nusrat? Is from the fact that it was very small study and he was only talking about single dose (4th dose) that not all got. Looking across doses, power much greater – and plenty of power to detect 50% increase immediately after vaccination (which is MUCH smaller than some suggest it should be).